

Older Adults' Experiences of Autonomy During COVID-19 Pandemic

Novia Nurain
Indiana University
Bloomington, Indiana, USA
nnurain@iu.edu

Clara Caldeira
Indiana University
Bloomington, Indiana, USA
cmarque@iu.edu

Kay Connelly
Indiana University
Bloomington, Indiana, USA
connelly@indiana.edu

ABSTRACT

A sense of autonomy is important to ensure a good quality of life for older adults. However, the recent COVID-19 pandemic challenged older adults' autonomy. The government has imposed rules and regulations to slow down the spread of the virus that reduced older adults' abilities to go outside. We conducted semi-structured interviews with 15 older adults living alone in community-dwellings to understand the influence of the pandemic on their experiences of autonomy. Our findings show that even though older adults experienced limited autonomy due to extrinsic (e.g., government regulation, influence of others) and intrinsic (e.g., risk perception) factors, they experienced new opportunities for autonomy through engagement in various self-chosen activities at home. The insights from the study pave ways for future research and design of aging in place technology focusing on providing a sense of autonomy.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

older adults, autonomy, pandemic

ACM Reference Format:

Novia Nurain, Clara Caldeira, and Kay Connelly. 2021. Older Adults' Experiences of Autonomy During COVID-19 Pandemic. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3411763.3451674>

1 INTRODUCTION

Around 8.5 percent of the world's population is aged 65 and over [8]. This percentage is projected to rise by nearly 17 percent of the world's population by the end of the year 2050. Ensuring the "quality of life" to this large population of older adults has become a prime concern for governments, society, and researchers. Older adults feel that the sense of autonomy is important to ensure good quality of life [13].

According to self-determination theory (SDT), the satisfaction of the need for autonomy, together with the need for relatedness

and competence, are essential for well-being and psychological growth [17]. However, the level of satisfaction, and the experience of autonomy are influenced by a variety of extrinsic and intrinsic factors such as age, life transitions, the context of living, pandemics, and life disruptions [9, 19]. We need to understand the influence of these factors on the experience of autonomy, as autonomy is an important factor for user experience in technology designed for older adults. In this paper, we explore the influence of pandemic on older adults' autonomy.

The recent outbreak of COVID-19 pandemic has affected peoples' lives and placed new challenges as everyone was supposed to maintain social isolation. However, for the older adults this is particularly challenging because they have higher risk [14], use less technology, and rely more on in-person services and interactions in comparison with other age groups. In addition to the above-mentioned challenges older adults may also face challenges satisfying their need for autonomy as the government imposed a variety of rules and regulations to slow down the spread of the virus. There exist a few research studies [6, 20] concerning the pandemic and its influence on older adults' autonomy. The research study presented in [20] found that the satisfaction of the need for autonomy was not consistently correlated to the well-being during the COVID-19 pandemic. The qualitative study presented in [6] found that older adults felt limited autonomy due to governmental restrictions during the pandemic, however, their autonomy satisfaction did not lower. These findings need further attention. Therefore, in this paper, we aim to answer the following research question: how is the experience of autonomy influenced by COVID-19 pandemic for older adults living alone at community-dwellings?

To answer the research question, we conducted semi-structured interviews with 15 older adults living alone at community-dwellings. Through our study, we found that on the one hand, older adult participants experience limited autonomy due to different extrinsic and intrinsic factors (for example, government guidelines and the influence of others) invoked by the pandemic. On the other hand, older adults experience new opportunities for autonomy while being stuck at home (for example, indulge in self-chosen activities at home). Our findings also uncovered the interplay between the need for autonomy and the need for safety that brewed tension around accepting assistance from others. Our findings contribute to HCI research by discussing older adults' experiences with autonomy and implications for aging in place technology research and design.

2 BACKGROUND

Autonomy is touted to be a vital entity of basic psychological needs. According to self-determination theory (SDT), autonomy together

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CHI '21 Extended Abstracts, May 8–13, 2021, Yokohama, Japan

© 2021 Association for Computing Machinery.

ACM ISBN 978-1-4503-8095-9/21/05...\$15.00

<https://doi.org/10.1145/3411763.3451674>

with other psychological needs, i.e., relatedness and competence, when satisfied, would promote well-being and psychological growth [5, 16]. We may define autonomy as having a sense of choice and volition in the regulation of behavior [17]. Webster et al., define autonomy as a “self-governing state” and “self-directing freedom” [10]. There are four factors integral to autonomy: 1) being free from the controlling influence of others, 2) being free from limitations that prevent meaningful choice, 3) being free from inadequate understanding, and 4) being able to freely act in accordance with a self-chosen plan [1].

Autonomy is essential across one’s life. Even though the need for autonomy is universal, both the level of autonomy and the level of satisfaction with autonomy seems to change throughout life [13]. The experience of autonomy may be threatened or inhibited by a variety of circumstances and factors, real or symbolic, intrinsic, or extrinsic to the person. Age is a common factor that influences the notion of autonomy. It is found that older adults experience a higher satisfaction of the need for autonomy compared to their younger selves [9]. There exists a growing body of evidence in the literature that suggests that older people themselves feel that autonomy and independence are important to good quality of life [12, 13].

Aging in place technology aims to extend independence for older adults [11]. By providing assistive services such as fall detection, such systems could allow older adults to delay the need to move in with caregivers or transition into an assistive living facility. Technologies for older adults and caregivers can also impact perceived autonomy, both positively and negatively [21], and this perception influences their use and experience with this kind of technology. For example, systems whose design limits perceived autonomy are often rejected or abandoned by older adults [3].

In addition, older adults’ need for autonomy may also alter due to changes in life situations such as life transition and life disruption [9, 18]. Life transition may be triggered by retirement and life disruption may be triggered by deaths, pandemics, and health issues. Context of living also influences the experience of autonomy of older adults. Older adults living in the home may experience autonomy in a different way in comparison to older adults living in assisted living facilities with more rules and restrictions. Most of the existing studies focus on strengthening the sense of autonomy of older adults living in nursing care facilities [4, 15]. A better understanding of the need for autonomy of older adults living at home is necessary to support the design of aging in place interventions aimed to provide a sense of autonomy. Furthermore, it is also necessary to understand how the sense of autonomy in such living conditions is threatened or inhibited by circumstances such as the COVID-19 pandemic. Therefore, in this study, we strive to understand the influence of the COVID-19 pandemic on the autonomy of older adults living alone at home during the pandemic.

3 METHOD

To investigate how older adults’ experience of autonomy is influenced by the COVID-19 pandemic, we conducted a qualitative study with older adults living in community dwellings. The study was conducted in a single state in the United States (US) Midwest. The study protocol was approved by the university institutional review board. We recruited older adults from May to early June

2020 through snowball sampling, flyers, and mailing lists associated with the local university, local organizations, and communities. For the study, we recruited 15 older adults and all of them are women. Their age ranges from 66 to 81, with an average age of 74 years old (Table 1). All of our participants have been living alone in the local community dwellings for about 10 or more years and they have a strong social connection with the community. As detailed in Table 1, 13 participants were retired, 5 participants involved in different part-time jobs, and 6 participants indulged in a variety of volunteer activities before the pandemic. During the pandemic, only 3 participants were involved in part-time jobs and volunteer activities.

We conducted semi-structured interviews from mid June to July 2020. The interview questions included the experiences and influences of the pandemic on the daily lives of the older adults. For example, we asked them how are they running errands, i.e., buying groceries during the lockdowns or stay-at-home orders. We also asked about how are they using online delivery services and other technologies during this time. The first author conducted the interviews. To ensure the safety of the older adult participants during the COVID-19 pandemic all the interviews were conducted remotely via Zoom call. Each interview lasted for about 45 to 60 minutes. Each participant was compensated with a \$10 Amazon electronic gift card for their time.

All interviews were recorded and transcribed for analysis. We then conducted a thematic analysis [7] following thematic analysis guidelines [2]. According to the guideline, the first author conducted an iterative open-coding analysis to get familiarized with the transcribed data and then followed inductive analysis to develop several codes regarding older adults’ experiences of autonomy during the pandemic, for example, opportunities, requirements, barriers to autonomy. All the authors had discussed the codes through synchronous meetings to organize them into emerging themes. In this paper, we present preliminary results from this analysis.

4 FINDINGS

Throughout the life course, the sense of autonomy might alter due to situations such as life transitions (for example, retirement) or life disruptions (for example, crisis, death of close ones, etc.). The recent COVID-19 pandemic can be viewed to be one of such life disruptions that challenged people’s autonomy due to the imposed government rules to slow down the spread. Participant OA2 mentioned: “I can’t go anywhere unless it’s to get food or go to the doctor or something like that. So I’m much more confined to my living space and to my very small neighborhood.”

On the one hand, our participants indicated that experienced limited autonomy due to different extrinsic and intrinsic factors invoked by the pandemic, such as governmental guidelines, the influence of others, inflexibility of the situation, and risk perception. On the other hand, the downtime being stuck at home allows them to explore a new dimension of autonomy through self-exploration. In this section, we begin by describing the perceived opportunities and requirements of autonomy by older adults during the pandemic. Next, we look into the factors that hinder their autonomy and explore the new opportunities of autonomy. Finally, we look into

Table 1: Demographics of the participants

Id	Gender	Age	Race	Education	Occupation before COVID-19	Occupation during COVID-19
OA1	F	71	White	Bachelor's Degree	Retired	Part-time
OA2	F	80	White	Doctorate Degree	Retired, Volunteer	Retired, Volunteer
OA3	F	73	White	Bachelor's Degree	Retired	Retired
OA4	F	74	White	Bachelor's Degree	Retired, Part-time, Volunteer	Retired
OA5	F	75	White	Bachelor's Degree	Retired	Retired
OA6	F	78	White	Doctorate Degree	Part-time	Part-time
OA7	F	79	White	Master's Degree	Retired, Part-time, Volunteer	Retired
OA8	F	74	White	Doctorate Degree	Retired, Volunteer	Retired, Volunteer
OA9	F	78	White	Bachelor's Degree	Retired	Retired
OA10	F	81	White	High School Diploma	Retired	Retired
OA11	F	69	White	Master's Degree	Part-time, Volunteer	Part-time
OA12	F	68	White	Master's Degree	Retired, Part-time	Retired
OA13	F	73	White	Master's Degree	Retired, Volunteer	Retired, Volunteer
OA14	F	72	White	Master's Degree	Retired	Retired
OA15	F	66	White	Master's Degree	Retired	Retired

the brewed tension between needing social support (mainly tangible support) and needing the sense of autonomy.

4.1 Perceived Opportunities and Requirements of Autonomy

Our analysis of the interview data revealed several perceived opportunities that give the feeling of autonomy to the older adults living alone at home during the pandemic time. We found that being able to go to the grocery store and choose produce by themselves gave them a sense of autonomy. For example, OA2 mentioned: "I like to choose my things myself. For example, if I want to choose produce fresh vegetables fresh fruit, I want to choose it myself. I don't want someone else to choose for me."

The participants pointed out that they feel the need to have the freedom of making the decision about their produce rather than someone else make the decision for them. For instance, OA5 informed that she didn't like to use the local grocery delivery services via Instacart, Kroger, Walmart, etc. during the pandemic. She felt that usage of such delivery services made her lose the ability to make decisions to buy food items as she had to rely on someone to make such decisions.

Participants also mentioned that being able to do self-chosen activities they like to do gave them the opportunity to experience autonomy. OA5 mentioned that being able to do unplanned activities during the pandemic invoked a sense of limited autonomy.

Our participants mentioned that being retired from work allowed them to do self-chosen activities such as walking, gardening, meeting people, going to the library, going to bed, etc. at their convenience. They viewed that being retired allows them to have enough time when they are confined at home during the pandemic. Having enough time provides them the opportunity to be autonomous to do things that they are not able to do when they are at work. For example, OA9 mentioned: "I'm retired so that's exactly what I did for, oh, I think I'll go to the library or I think I'll call somebody and say, you know..."

The above-mentioned findings are aligned with the factors of autonomy mentioned in Section 2, more specifically the fourth factor, i.e., being able to act freely with a self-chosen plan. Our analysis also showed that having their own cars and being able to drive by themselves provided the older adults a sense of autonomy during the pandemic time. OA2 mentioned that even if she is 80

years old, she still drives. Another participant OA1 mentioned that having a car allows her to go to places whenever she wants to. It made her independent enough not to need any social support for local resources during the pandemic time.

Our analysis also unveiled several perceived requirements such as technological competence, finances, transportation, and decision making power that are needed in order to experience autonomy by the older adults during the pandemic.

4.1.1 Technological Competence. Participants mentioned the need of having technological competency became more relevant with the pandemic as a vital requirement to experience the opportunities of autonomy. Since getting help from others is harder during the pandemic, technological competency would allow the participants to receive informational support about different available services for senior citizens. For instance, OA5 mentioned: "I use my computer a lot. I'm pretty computer literate. Either I'm able to look things up a little bit easier than someone else who is not computer literate, who doesn't have computers or or whatever. So I know that there are a lot of services available for the senior citizens here in town."

4.1.2 Finances. The second requirement that has been mentioned by our participants is having enough financial resources during the pandemic. Financial resources are needed to feel autonomous as some other things that give a feeling of autonomy (for example, choosing foods and produce) are not possible without this requirement. OA5 mentioned, "I am fortunate enough to be able to afford to pay delivery costs."

4.1.3 Access to Transportation. Our participants emphasized having access to transportation as a requirement to experience autonomy during the pandemic. Having access to transportation allowed the participants to become more independent to fulfill their needs rather than being home-bound. OA10 mentioned: "During this pandemic time I go get my own (grocery and food products). I have car." Besides, access to transportation provided the opportunity to do tasks independently: "I have a car, which is my best friend. When it goes, I really don't want to live anymore, but when I can't drive and do stuff for myself (talking about when she was not able be independent). I really don't want to live anymore." (OA7)

4.1.4 Decision Making Power. Our analysis showed that the participants considered that having decision making power contributed

a large part to their feeling of autonomy. Participants mentioned having control over the decision of social interactions during the pandemic. For example, to experience autonomy during the pandemic the participants liked the freedom to decide how much they want to be around other people. OA15 mentioned: “everybody’s decided on their own of how, what they’re comfortable with and how much they want to be around other people or not.”

In addition, participants also wanted to have the decision making power over the selection of their food purchases during the pandemic. OA1 mentioned: “just wanting to make my own selections. I get to the store and look because I’d see if something’s on sales and then knew if something looks good, or strikes, my interest and that’s what I’ll get.”

In summary, we found that requirements for autonomy such as having technological competence and decision making power become more prominent during the pandemic scenario. It revealed a shift to the notion of being independent and autonomous as older adults and this shift is triggered by the COVID-19 pandemic.

4.2 Factors Hindering the Feelings of Autonomy

Individual’s life has been infiltrated by COVID-19 pandemic in many ways. People’s autonomy is one of the aspects of life that has been most challenged by the pandemic. Our findings uncovered the influence of several factors that hindered the sense of autonomy of older adults during the pandemic.

4.2.1 Government Rules and Regulations. To slow down the spread of the virus and to ensure the safety of the people, governments had imposed rules and regulations. The influence of the government-imposed rules and regulation hindered the autonomy of older adults to some extent by restricting the opportunities of autonomy (for example, doing self-chosen activities). Participants mentioned experiencing limited autonomy due to different governmental guidelines and restrictions. OA6 mentioned: “well they (Government) didn’t say could not take walks. So I would take walks, the streets were empty, so I take walks.”

4.2.2 Lack of Control. In addition to external rules and regulation, the inflexibility and uncertainty of the pandemic situation influence the feeling of autonomy. Participants mentioned that due to the unique nature of the pandemic, the rules, regulations, and protective measures against the virus changed all the time. The lack of control over the situation hindered the above-mentioned decision making power requirement of autonomy. For instance, OA9 mentioned: “I think it’s the inflexibility of it that I can’t decide what I want to do and then go do it I think the biggest thing is not not just being able to do.”

Participants also mentioned that the uncertainty of not being able to see one’s children, grandchildren, or friends was described as a big limitation to one’s feeling of autonomy. OA14 specified: “I have two children, and my son is 34 and he lives in the Netherlands and I don’t even know when I get to see my son again and in real life.”

4.2.3 Influence of Others. Our analysis showed that the influence of other people hindered the older adults feeling of autonomy by exerting pressure. For instance, OA10 mentioned about her friend

whose autonomy of going to places was limited by the intrusion of her family members: “one of the kids don’t want her going anywhere because she might get it (virus). They’re kind of taken over and boss her around on what she can do or not.” Participants also mentioned that the protectiveness of family members and friends due to their age limit their autonomy to some extent. OA2 who is 80 years old, mentioned that her daughter is very protective of her and didn’t allow her to go to grocery stores by herself. Another participant OA9 also mentioned such instance of protectiveness of others: “My sons and my one family friend decided that I should not go to the grocery store [...] they did not want me going to the grocery store, even though I’m healthy. I don’t have any reason not to go, they just didn’t think it was safe.”

4.2.4 Perceived Risk of Contamination. Participants mentioned that the risk of being contaminated by the virus also influenced their autonomy. In such cases, the participants themselves limited their requirements of autonomy to ensure their safety against the virus during the pandemic. For instance, OA13 mentioned not being able to go back to the recreational centers due to the perceived risk of being affected by the coronavirus: “I don’t know if I’m you know even going to be able to go back to the Y, the swimming pool . . . they had camp for for groups of children and children just they are carriers of COVID.” In summary, our findings exposed several factors that hinder the feeling of autonomy of older adults during the pandemic. These factors sometimes are triggered by external entities such as the government, family members, or friends. On the other hand, in some instances, intrinsic factors of the older adults are responsible to limit their autonomy during the pandemic.

4.3 Tension Around Accepting Assistance and Autonomy

Our analysis revealed the presence of tension around accepting assistance and autonomy. On the one hand, participants mentioned that they felt autonomous when it comes to choosing produce and doing errands by themselves. Hence, they decide not to accept assistance or social support from others. For instance, OA12 mentioned: “our neighborhood would always have a little pitch in every week, and so I have some friends [...] and somebody from that group told me that if I needed her to do any errands for me. She would be happy to do, but I haven’t needed to.”

Our analysis also showed that even though there were strict restrictions caused by the COVID-19 pandemic limiting older adults’ experience of autonomy, participants still experienced a feeling of autonomy in some areas of their lives. Participants mentioned that the downtime during the pandemic allows more available time to do a variety of self-chosen activities inside the home. They were free in their decisions about what to do when having leisure time. Our findings showed that participants indulged in leisure activities such as making saddles, cards, house projects, gardening, etc. For instance, OA8 mentioned: “I have my garden and so gardening takes up some of my time now (during the pandemic)”. Participants also found ways to enjoy “me time” that gave a feeling of autonomy inside the home during the pandemic time. OA14 mentioned: “I get some sparkling water and I go down to my guest room and I have a couch. It has lots of bolsters and pillows and I go, and both my

animals, follow me and they sit on top of me. And it's like, and it's like cozy time. So for about an hour we watch something . . ."

On the other hand, participants also mentioned compromising their need for autonomy to reduce the risk of getting contaminated with the virus. For instance, OA11 mentioned: "Previously, like I said, I like going to the grocery store and I had thought Oh, I don't want somebody else picking out my avocados [...], but then the risk felt higher than it was like okay well I can deal with somebody can choose my avocado in instead of, you know, risking getting sick."

We found a brewed tension between needing social support or needing to have autonomy became more prevalent for older adults due to the COVID-19 pandemic. Older adults are more vulnerable to the virus, hence, the need for autonomy is constrained by the need for safety in some cases. The risk perception influenced more while deciding on accepting assistance from others.

5 DISCUSSION

Our preliminary findings align with and further validate the factors integral to autonomy (mentioned in Section 2). We found several requirements, such as decision-making power, finance, and transportation. These requirements allow older adults to be free from limitations that prevent meaningful choice, which is an integral factor to autonomy. The findings also uncovered several factors such as government guidelines, lack of control over the pandemic situation, and risk perception that limit older adults' sense of autonomy by violating the factor of autonomy, i.e., being able to act freely following a self-chosen plan. We found that during pandemic in some cases older adults' self-chosen plans and choices are controlled by the influence of others. Older adults experienced a sense of limited autonomy as the COVID-19 pandemic perturbed the factors integral to the perception of autonomy.

During the COVID-19 pandemic, certain systems such as online grocery delivery services have become more prevalent to reduce risks of exposure to the virus. This kind of system should have been particularly useful for older adults, who are among the groups with the highest risks of experiencing severe complications of COVID-19 [14]. However, we found that older adults experienced frustrations with such systems. Older adults prefer to touch, feel, and explore the produce during grocery shopping. The limitation of the current online grocery systems is: these systems are not being able to provide the touch and feel experience. At the same time, older adults found it difficult to describe their perception about the quality of the produce (i.e., freshness) to the third party. A system supporting real-time video calling might have the potential to address these issues to some extent.

Next, relying on unknown people to make purchasing decisions for the grocery delivery was described as a risk factor during the pandemic time by our study participants. To address this issue, systems could be designed around trust-building between the older adults and the people who will buy the produce for them. For example, the assignment of the same person who will deliver to particular older adults over time and thus will be able to understand their preferences. Furthermore, our findings also indicated a sense of disempowering experienced by our study participants. This issue highlights a need to rethink how these online delivery systems can

be re-designed to improve the experience of autonomy of older users by providing them a sense of agency.

Our findings uncovered tension around accepting assistance from others during the pandemic. We could see the presence of similar tension in older adults' experiences with aging in place technology, as they reject systems that reduce their sense of independence (e.g., due to aging-related stigma) [3]. Ideally, technology design should aim to reduce such tensions by eliminating the need for older adults to sacrifice some of their perceived autonomy to fulfill their needs. Our findings highlight opportunities for technology design to prevent negative impacts on older adults' sense of independence. By aiming to incorporate factors that help with autonomy (e.g., decision making) and reducing factors that limit autonomy (e.g., rule setting by caregivers), designers can adapt aging in place systems to strengthen a sense of independence among older adult users. For example, designing systems or tools to create opportunities for online volunteer activities to maintain their independence.

Finally, our findings indicated that apart from limited autonomy during the pandemic, older adults experienced new opportunities for autonomy through self-care. Hence, technology design should also focus on developing services or tools to support self-care and to empower older adults. For example, a system that promotes ways for older adults to enjoy themselves, such as connecting with arts and groups, learning a new hobby, and engaging in leisure activities.

6 LIMITATIONS

We tried to recruit participants using a variety of recruitment instruments. However, due to the pandemic and stay-at-home order, our sample is skewed towards participants being white, female, and living in the US. Most of our participants belong to either mid-SES or high SES. Hence, the findings might not be widely applicable to the broader population of older adults across different SES. For example, most of our participants used their transportation to go to grocery stores to sustain their decision-making power. On the other hand, older adults from low SES might use public transportation to go to grocery stores that might expose them to more risk factors during the pandemic. Thus, maintaining a sense of autonomy will become more challenging for low SES older adults.

7 CONCLUSION

In this paper, we present the preliminary findings of a qualitative study investigating the experiences of autonomy of older adults living alone in community-dwellings. Our findings suggest that older adults experienced limited autonomy due to a variety of factors, such as government guidelines, the influence of others, and risk perception. While being confined at home during the COVID-19 pandemic, older adults explored new opportunities for autonomy through engagement in self-chosen activities. Based on these findings, we present several factors that impact perceived autonomy among older adults. These insights can inform future research and design to examine how to support older adults in achieving a sense of autonomy, accounting for the changed notion of autonomy due to the pandemic.

ACKNOWLEDGMENTS

We would like to thank all the participants for their time and for sharing their experiences with us. This work was supported in part by the Indiana University Center for Rural Engagement, the Indiana University Grand Challenge Precision Health Initiative, the National Science Foundation Grant No. 1629468, and the National Science Foundation under Grant No. 2030859 to the Computing Research Association for the CI Fellows Project.

REFERENCES

- [1] Tom L. Beauchamp and James F. Childress. 2001. Respect for autonomy. *Principles of biomedical ethics* 5 (2001), 57–112.
- [2] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative research in psychology* 3, 2 (2006), 77–101.
- [3] Clara Caldeira, Matthew Bietz, Marisol Vidauri, and Yunan Chen. 2017. Senior care for aging in place: Balancing assistance and independence. In *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing*. 1605–1617.
- [4] Sue Davies, Lorraine Ellis, and Sara Laker. 2000. Promoting autonomy and independence for older people within nursing practice: an observational study. *Journal of Clinical Nursing* 9, 1 (2000), 127–136.
- [5] Edward L. Deci and Richard M. Ryan. 2000. The "what" and "why" of goal pursuits: Human needs and the self-determination of behavior. *Psychological inquiry* 11, 4 (2000), 227–268.
- [6] Jannike Hachmeister. 2020. *A qualitative analysis of how older adults experience that their need for autonomy has changed throughout their life and during COVID-19 pandemic*. Master's thesis, University of Twente.
- [7] Beverley Hancock, Elizabeth Ockleford, and Kate Windridge. 2001. *An introduction to qualitative research*. Trent focus group.
- [8] Wan He, Daniel Goodkind, Paul R. Kowal, et al. 2016. An aging world: 2015. (2016).
- [9] Georg Henning, Andreas Stenling, Susanne Tafvelin, Isabelle Hansson, Marie Kivi, Boo Johansson, and Magnus Lindwall. 2019. Preretirement Work Motivation and Subsequent Retirement Adjustment: A Self-Determination Theory Perspective. *Work, Aging and Retirement* 5, 2 (2019), 189–203.
- [10] Inc Merriam-Webster. 1983. *9,000 Words: A Supplement to Webster's Third New International Dictionary*. Merriam-Webster.
- [11] Elizabeth D. Mynatt, Irfan Essa, and Wendy Rogers. 2000. Increasing the opportunities for aging in place. In *Proceedings on the 2000 conference on Universal Usability*. 65–71.
- [12] Anne EM Nyström and Kerstin M Segesten. 1994. On sources of powerlessness in nursing home life. *Journal of Advanced Nursing* 19, 1 (1994), 124–133.
- [13] Marjone Oleson, Carol Heading, Kathleen McGlynn, and Jo Ann Bistodeau. 1994. Quality of life in long-stay institutions in England: nurse and resident perceptions. *Journal of Advanced Nursing* 20, 1 (1994), 23–32.
- [14] World Health Organization et al. 2020. Coronavirus disease 2019 (COVID-19): situation report, 72. (2020).
- [15] Jari Pirhonen, Helinä Melkas, Arto Laitinen, and Satu Pekkarinen. 2020. Could robots strengthen the sense of autonomy of older people residing in assisted living facilities?—A future-oriented study. *Ethics and Information Technology* 22, 2 (2020), 151–162.
- [16] Richard M. Ryan and Edward L. Deci. 2000. The darker and brighter sides of human existence: Basic psychological needs as a unifying concept. *Psychological inquiry* 11, 4 (2000), 319–338.
- [17] Richard M. Ryan, Veronika Huta, and Edward L. Deci. 2008. Living well: A self-determination theory perspective on eudaimonia. *Journal of happiness studies* 9, 1 (2008), 139–170.
- [18] Richard M. Ryan and Jennifer G. La Guardia. 2000. What is being optimized?: Self-determination theory and basic psychological needs. (2000).
- [19] Carol D. Ryff. 1995. Psychological well-being in adult life. *Current directions in psychological science* 4, 4 (1995), 99–104.
- [20] Rui Sun, Alisa Balabanova, Claude Bajada, Yang Liu, Mariia Kriuchok, Silja-Riin Voolma, Mirna Đurić, Claude-Hélène Mayer, Maria Constantinou, Mariam Chichua, et al. 2020. Psychological wellbeing during the global COVID-19 outbreak. (2020).
- [21] John Vines, Stephen Lindsay, Gary W. Pritchard, Mabel Lie, David Greathead, Patrick Olivier, and Katie Brittain. 2013. Making family care work: dependence, privacy and remote home monitoring telecare systems. In *Proceedings of the 2013 ACM international joint conference on Pervasive and ubiquitous computing*. 607–616.